

nnUNet set up

1. Create folder for your project on Linux – Ubuntu
2. Open wsl

Following steps are conducted in wsl

3. Move to your folder (**cd ~/home/your_folder**)
4. Create python environment (**python3 -m venv environment_name**)
5. Activate python env (**source /home/folder_name/environment_name/bin/activate**)
6. Check if CUDA runs correctly (**nvidia-smi**)
7. Installation of torch torchvision torchaudio (**pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu126**)
8. Check if GPU is recognized → “True” (**python -c "import torch; print(torch.cuda.is_available())"**)
9. Installation of nnUNet (**pip install nnunetv2**)
10. Create folder structure for nnUNet
mkdir -p nnUNet_raw (contains your dataset which will be used for training)
mkdir -p nnUNet_preprocessed (contains data from preprocessing, e.g. training parameters)
mkdir -p nnUNet_results (contains results after training a nnUNet model on your dataset)
11. Export/define paths; nnUNet needs to know where the data is
export nnUNet_raw="\$HOME/BHS_project/nnUNet_raw"
export nnUNet_preprocessed="\$HOME/BHS_project/nnUNet_preprocessed"
export nnUNet_results="\$HOME/BHS_project/nnUNet_results"
12. Check if paths are defined correctly (not necessary) → returns your paths to raw, preprocessed and results folders
echo \$nnUNet_raw
echo \$nnUNet_preprocessed
echo \$nnUNet_results
13. Start preprocessing of dataset → results saved in nnUNet_preprocessed
nnUNetv2_plan_and_preprocess -d 1 --verify_dataset_integrity
14. Start training of model
nnUNetv2_train 1 2d 0 (0-4 at the end = fold number)
nnUNetv2_train 1 2d 1
...
nnUNetv2_train 1 2d 4
15. Start prediction on new data
**nnUNetv2_predict **

**-i nnUNet_raw/Dataset001_name/imagesTs **
**-o predictions **
**-d 1 ** (dataset used for training)
**-c 2d ** (2d nnUNet model)
-f 0 (fold number, five folds: -f 0 1 2 3 4)

Dataset structure

(link to dataset download: <https://filesender.aco.net/?s=download&token=5bf311b7-664f-4fa7-a33a-86b8db8bfdaf>)

```
nnUNet_raw/
├─ Dataset001_name/                                (Dataset001 = 1 used in steps 13-15)
│  └─ dataset.json
│  └─ imagesTr/
│     │ └─ img_001_0000.png
│     │ └─ img_002_0000.png
│     │ └─ ...
│     └─ labelsTr/
│        │ └─ img_001.png
│        │ └─ img_002.png
│        │ └─ ...
│        └─ imagesTs/ (optional)
│           └─ img_101_0000.png
│           └─ ...
```